NS Ticketing System - Documentation

1. Introduction

The NS Ticketing System is an enterprise-level IT Service Management (ITSM) application built using Microsoft Power Platform.
It is designed to manage IT operations, track incidents, monitor SLA performance, and provide reporting insights.

2. Technologies Used

Power Apps (Canvas App)
Power Automate (Flows & Notifications)
SharePoint Online (Database)
Microsoft Entra ID (Authentication & Roles)
Microsoft Graph API (Get Custom Attribute From Entra)
Office365Users Connector (AD)
Outlook Connector

3. System Modules

- Ticket Management
- User Portal (My Tickets)
- IT Dashboard (Reports)
- User Search
- IT Team Directory
- Asset Tracking
- Consumables Tracking

4. SharePoint Structure

Main List: TicketsRequests
Includes:
TicketID, Affected User, Assigned To, Area, Category, Site, Status,
Priority, Severity, SLA Target, Resolved Time, SLA Breached, Ticket Type

AssetsMovements List:
Device, From Location, To Location, Status, Approval

5. Screens Overview

Home Screen: Navigation hub
Create Ticket Screen: Ticket creation by IT
My Tickets Screen: User tickets
Ticket Details Screen: Manage ticket
IT Dashboard: Reports & Charts
User Search Screen: Search tickets by user
IT Team Screen: Staff directory
Assets Screen: Device movement tracking
Consumables Screen: Consumables tracking

6. Role-Based Access

Controlled using variable: varIsIT

IT Users:
- Add Ticket
- All Other Options

Normal Users:
- View their own tickets
- View IT team information
- Track ticket status
- Move Assets

7. Email Notification Workflow

Triggered when a ticket is created.

Recipients:
- Area Manager
- User Manager
- Infrastructure Manager
- Service Desk
- Assigned Engineer
-Affected User

Purpose:
- Fast communication
- SLA compliance
- Visibility

8. SLA Management

Fields:
- SLA Target
- Resolved Time
- SLA Breached

Logic:
If Resolved Time > SLA Target → SLA Breached = Yes

9. Reporting Dashboard

Reports:
- Tickets by Status
- Tickets by Site
- Tickets by Type
- Engineer Workload
- Top 5 Users

These reports support decision making and workload analysis.

10. Asset Tracking Module

Tracks device movement between locations.

Workflow:
1. Request created
2. Approval required
3. Status updated (Approved/Rejected)

Purpose:
- Asset visibility
- Control device movement

11. Architecture

User
↓
Power Apps
↓
SharePoint
↓
Power Automate
↓
Email Notifications
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Dashboard

12. Security

- Entra ID Authentication
- Role-based UI
- SharePoint permissions

13. App Objectives

- Centralize all IT incidents and service requests
- Improve IT response and resolution times
- Track ticket lifecycle and SLA compliance
- Provide real-time operational reporting
- Improve communication between IT and employees

14. Programming Languages & Scripting

- Power Fx – Application logic inside Power Apps
- JSON – Power Automate configuration
- HTML – Email formatting
- OData – SharePoint queries and filtering
- REST API – Integration with Microsoft services

15. Ticket Data Model

- Ticket ID – Unique ticket identifier
- Affected User – Person who reported the issue
- NSID – Unique number for the end user
- Area – IT service area (Infrastructure, Network, etc.)
- Assigned To – IT engineer responsible for the ticket
- Category – Issue classification
- Date – Ticket creation date
- Department – User department
- Device – Related device
- Issue Details – Description of the issue
- Severity – Incident severity level
- Priority – Ticket priority
- Impact – Business impact level
- Status – Ticket status
- Solution Steps – Resolution steps
- Proactive Action – Preventive actions
- Root Cause Category – Root cause analysis
- Other Affected Users – Additional impacted users
- Attachments – Supporting files

16. Ticket Data Model

Issue Reported

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Ticket Created

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Email Notification Sent

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Ticket Assigned

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Issue Investigation

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Solution Implemented

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Solution Documented

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Ticket Closed

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SLA Evaluation

17. Application Screens

The application is structured into key functional areas to ensure smooth navigation and efficient operations. The **Home Dashboard** acts as the main entry point, offering quick access to all modules and system insights. Service Desk Team can report users issues through the **Ticket Creation** interface, while the **My Tickets** section allows them to track their requests and monitor progress.
Detailed information for each request is available within the **Ticket Details** view.
For performance monitoring, the **Reports Dashboard** provides analytical insights and key metrics. The **IT Team** section introduces system users to the support staff and their roles.
Administrative control is handled through the **All Tickets** view, enabling full ticket management across the system. Additionally, the system includes dedicated modules for resource handling, such as **IT Assets** for managing equipment and transfers, and **IT Consumables** for tracking and issuing consumable items.

18. Reporting & Analytics

The dashboard includes multiple visual reports:
• Tickets by Area
• Tickets by Status
• Tickets by Month
• Tickets by Year
• SLA Breached Tickets
• Resolution Time Metrics

19. Future Enhancements

• Power BI Advanced Analytics
• Microsoft Teams Integration
• Mobile Optimization

20. Conclusion

The NS Ticketing System provides a centralized, scalable solution for managing IT support operations.

By leveraging Microsoft Power Platform technologies, the organization benefits from automation, visibility, and improved service delivery.